

Vendor Performance Analytics — Final Insights & Recommendations

Executive Summary

This project analyzed vendor performance across **131 vendors and 12,261 brand relationships**, using 2024 transaction data spanning \$450.88M in sales, \$320.91M in purchases, and \$1.64M in freight costs. The goal was to move purchasing decisions from relationship-based judgment to data-backed evidence.

Bottom line: Vendor size does not predict profitability — decisions should be based on each vendor's actual margin, not assumed scale advantages. Seven vendors show severe, isolated losses warranting immediate review. The business is overall healthy, running a 28.83% blended gross margin.

1. Business Problem Recap

Purchasing decisions (who to order more from, who to renegotiate freight with, who to drop) were being made without a clear, consolidated view of vendor profitability, inventory efficiency, or freight cost impact. This project built that view end-to-end: SQL Server for governed data infrastructure, Python for validated KPI engineering and statistical testing, and Power BI for an ongoing monitoring dashboard.

2. Headline Numbers

Metric	Value
Total Sales	\$450.88M
Total Purchase Cost	\$320.91M

Total Gross Profit	\$129.97M
Overall Margin	28.83%
Total Freight Cost	\$1.64M (0.51% of purchase cost)
Active Vendors	131
Brands Tracked	12,261

3. Key Finding 1 — Vendor Size Does Not Predict Margin

We tested whether higher-sales-volume vendors deliver better profit margins, using two independent statistical tests at the correct unit of analysis (vendor-level, n=67 high-volume vs. n=61 low-volume):

Test	Result	P-value
Welch's t-test (mean-based)	Not significant	0.093
Mann-Whitney U (outlier-robust, median-based)	Not significant	0.101

Median margins were close (27.4% high-volume vs. 23.6% low-volume vendors) — the earlier appearance of a gap was driven by a handful of extreme outlier vendors, not a genuine volume-based pattern.

Implication: Do not default to favoring large vendors in negotiations or allocation decisions on the assumption that scale equals better pricing. Evaluate every vendor — large or small — on its own actual margin performance.

4. Key Finding 2 — Top Performing Vendors (by Gross Profit)

Vendor	Total Gross Profit	Overall Margin %
Diageo North America Inc	\$17.78M	25.87%
Martignetti Companies	\$13.19M	32.14%
Constellation Brands Inc	\$8.90M	36.35%
Pernod Ricard USA	\$8.16M	25.27%
Jim Beam Brands Company	\$7.70M	24.14%

These 5 vendors alone generate ~\$55.7M, roughly 43% of total gross profit. They represent the highest-priority relationships to maintain and, where possible, expand.

Note: Martignetti Companies initially appeared unprofitable under a naive percentage-averaging calculation — corrected to a dollar-weighted margin methodology, it is confirmed as a genuinely strong, top-5 vendor. This distinction matters for any future re-analysis: **always weight vendor margin by dollars, not by averaging brand-level percentages.**

5. Key Finding 3 — Seven Vendors Require Immediate Review

These vendors show severe negative margins that are not explained by market segment or sales volume — each is an isolated case:

Vendor	Gross Profit	Margin %
Truett Hurst	-\$221.65	-1,478.65%

Ira Goldman and Williams, LLP	-\$4,365.30	-344.92%
Highland Wine Merchants LLC	-\$3,966.64	-258.64%
Vineyard Brands LLC	-\$6,627.55	-153.27%
Uncorked	-\$1,841.93	-163.82%
Black Cove Beverages	-\$8,208.19	-131.19%
Loyal Dog Winery	-\$1,209.54	-108.84%

Combined direct dollar loss is modest (~\$26.4K), but the **pattern is the real signal**: each of these vendors is losing money on effectively every unit sold, independent of scale. This typically points to pricing or purchase-cost issues specific to that vendor relationship, not a market-wide trend.

Recommendation: Route these 7 vendors to Purchasing for individual pricing renegotiation, or evaluate discontinuing the relationship if renegotiation isn't viable.

6. Key Finding 4 — Freight Cost Is Not a Meaningful Margin Driver, With Two Exceptions

Across most vendors, freight cost sits in a narrow, healthy band of **0.46%–0.64%** of purchase dollars — too little variation to explain margin differences.

Two outliers stand apart: Vineyard Brands LLC (55.43%) and Southern Glaziers W&S of NE (46.60%) — freight costs on these two are dramatically higher than every other vendor. Given Vineyard Brands is also one of the 7 flagged loss-making vendors, this may be a contributing (or compounding) factor.

Recommendation: Investigate these two vendors' invoice records directly — confirm whether this reflects genuinely small/costly shipments, a data entry issue, or a real freight-negotiation gap worth addressing.

7. Data & Methodology Notes

- All 6 source tables (begin/end inventory, purchase prices, purchases, sales, vendor invoices) were loaded into SQL Server, verified for row-count accuracy, type consistency, and null patterns before any analysis began.
- NULLs in activity fields were confirmed to represent **genuine zero activity** (not data errors) via spot-check joins before being filled with 0.
- Two calculation bugs were caught and corrected during the analysis:
 1. **Freight Cost % grain mismatch** — initially divided vendor-level freight by brand-level purchase dollars, producing invalid results as high as 17,000,000%. Fixed by recalculating at the correct vendor grain.
 2. **Margin ranking bias (Simpson's Paradox pattern)** — naive averaging of brand-level percentages misrepresented some genuinely strong vendors as unprofitable. Fixed by switching to dollar-weighted margin ($\Sigma \text{Profit} / \Sigma \text{Sales}$).
- Ratio-based KPIs (Stock Turnover, Profit Margin %) are flagged as unreliable for individual brands with very low sales volume (<\$500) or near-zero average inventory (<2 units) — these brands remain in all totals/aggregates but are excluded from ratio-based rankings to avoid distortion from small denominators.

8. Recommendations Summary

Priority	Action	Owner
High	Individually review pricing/terms with the 7 flagged loss-making vendors	Purchasing
High	Investigate freight anomaly at Vineyard Brands LLC and Southern Glazers W&S of NE	Purchasing / Logistics
Medium	Maintain and prioritize relationships with top-5 profit vendors (Diageo, Martignetti, Constellation, Pernod Ricard, Jim Beam)	Purchasing / Leadership

Medium	Stop using vendor size/volume as a proxy for expected margin in future negotiations	Purchasing / Leadership
Ongoing	Use the Power BI dashboard for continuous vendor monitoring rather than point-in-time review	Finance / Ops

9. Limitations & Next Steps

- This analysis covers a single calendar year (2024); trend analysis across multiple years would strengthen confidence in vendor patterns (e.g. is a flagged vendor's loss a one-time event or recurring?).
- Freight cost is invoiced at the vendor level, not itemized per brand/shipment — a more granular freight allocation (if available) could sharpen per-brand profitability further.
- Statistical tests here establish that vendor *size* doesn't predict margin — a natural next question is what *does* (e.g. product category, region, contract terms), which would require additional data not currently captured in this dataset.